POSTER BOARD: 1251

Does Electrocution Improve Roller-Crimper Cover Crop Termination?

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Organic growers can't use herbicides to kill cover crops, and the crimper alone often leaves survivors that come back.

Cereal rye · soft dough | Zasso EH60 · 2.5 m | College Station, TX · 2026

SETTINGS · 3.2 km h^{-1} · ONE PASS
Electrocution: 2.5 m · 255 V phase-to-neutral (generator), 40–45 A per phase · 138–155 MJ ha^{-1}



Electrocution (EWC)



Roller crimper (RC)

TREATMENTS · SAME-DAY PASSES

— Control ● EWC ■ RC ▲ RC→EWC ◆ EWC→RC

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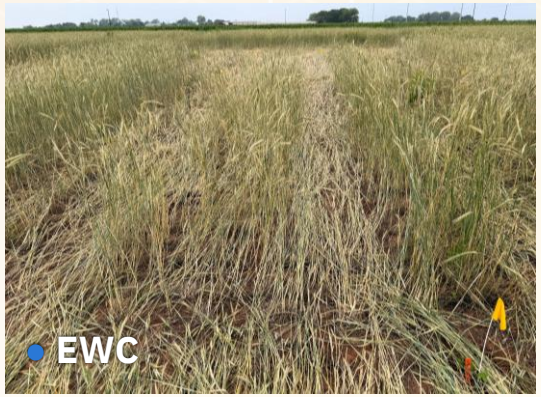
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RESULTS · 28 DAYS AFTER TREATMENT

EWC and the crimper together, in either order, terminated 80–90% of the rye

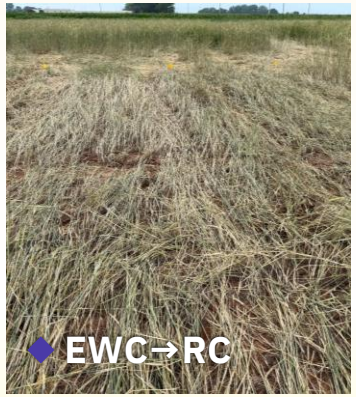
Field plots, 15 days after treatment



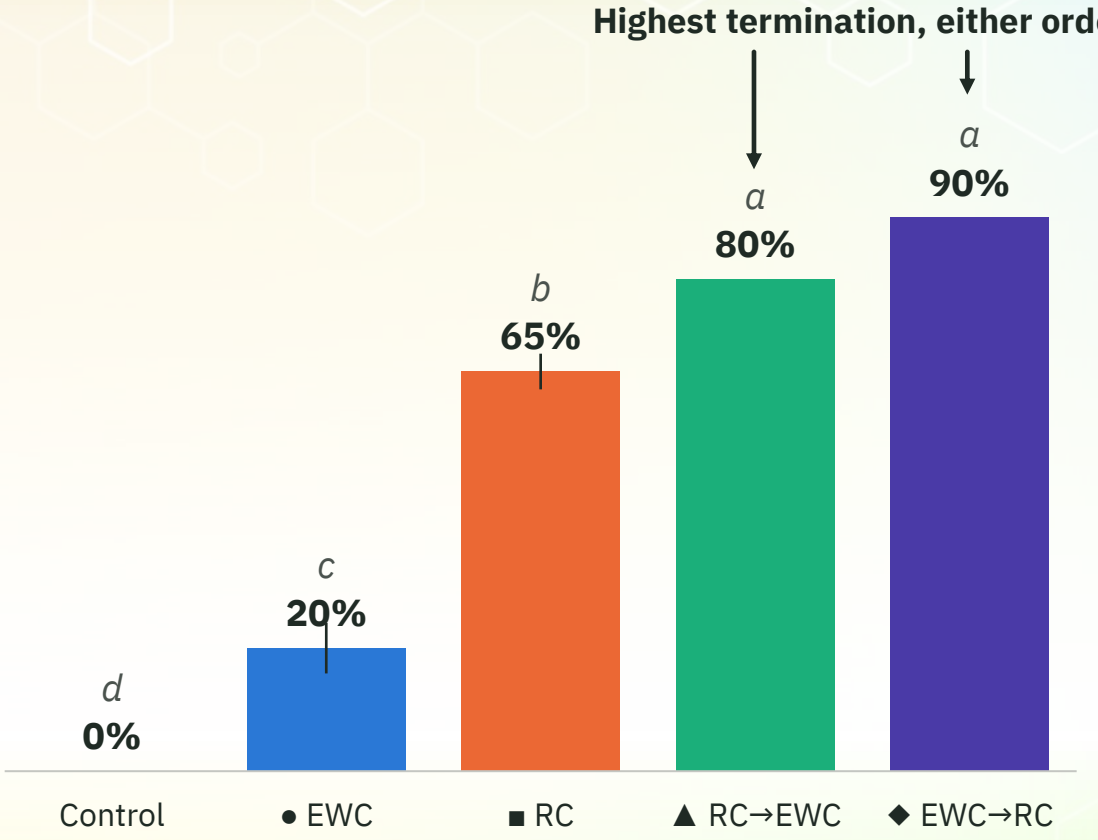








Termination



Cereal rye at soft dough (Zadoks 85), College Station, TX, 2026; RCBD, n = 4. ANOVA P < 0.001; mean ± SE; same letter = no difference (Tukey HSD, α = 0.05).
Initial biomass ≈ 615 g m^{-2} in all plots (P = 0.75). EWC: Zasso EH60, 2.5 m applicator, 3.2 km h^{-1}, ≈ 138–155 MJ ha^{-1} (upper bound). Sequences applied the same day.

RECOMMENDATIONS

Pair electrocution with the crimper; the order does not matter

80–90%

termination with EWC→RC or RC→EWC

6–11% of the stand survived and standing biomass fell 57–62%.

79%

of the stand survived EWC alone

Top-kill, not termination: biomass matched the untreated control, even at 138–155 MJ ha^{-1}.

34% → 6–11%

survival, crimper alone vs. crimper plus EWC

A same-day EWC pass closes the gap the crimper leaves.

TAKEAWAY RECOMMENDATION

Follow roller-crimping with a same-day electrocution (EWC) pass.

The crimper alone left a third of the rye alive; together they finished the job in either order.

NEXT STEPS Earlier growth stages · Other cover crop species · Energy dose · Precipitation records · Cash-crop yield

Preliminary: one site-year, 4 replications. Sequence order did not differ (termination P = 0.08; survival P = 0.84; biomass P = 0.69).

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